

SAMPLE Examination in the School of Computer and
Mathematical Sciences

STATS 7022

**SAMPLE EXAM
Data Science PG**

Instructions:

- Attempt all questions.
- Start your answer to each question on a new page.
- Clearly label your answers by question and part.
- You must show working in your answers to every question. Full marks will not be awarded without supporting working.
- This is a closed book exam. You are only permitted access to the materials listed below.

Permitted Materials:

- A scientific calculator or a basic/four function calculator. Calculators with graphics, remote communication, or CAS capabilities are NOT permitted.
- One hard copy of an English language translation dictionary.

Duration:

Writing Time: 120 minutes

This exam has 4 questions for a total of 80 marks.

Question:	1	2	3	4	Total
Marks	20	20	20	20	80

**DO NOT OPEN THIS BOOKLET OR COMMENCE WRITING
UNTIL INSTRUCTED TO DO SO. STOP WRITING
IMMEDIATELY WHEN INSTRUCTED.**

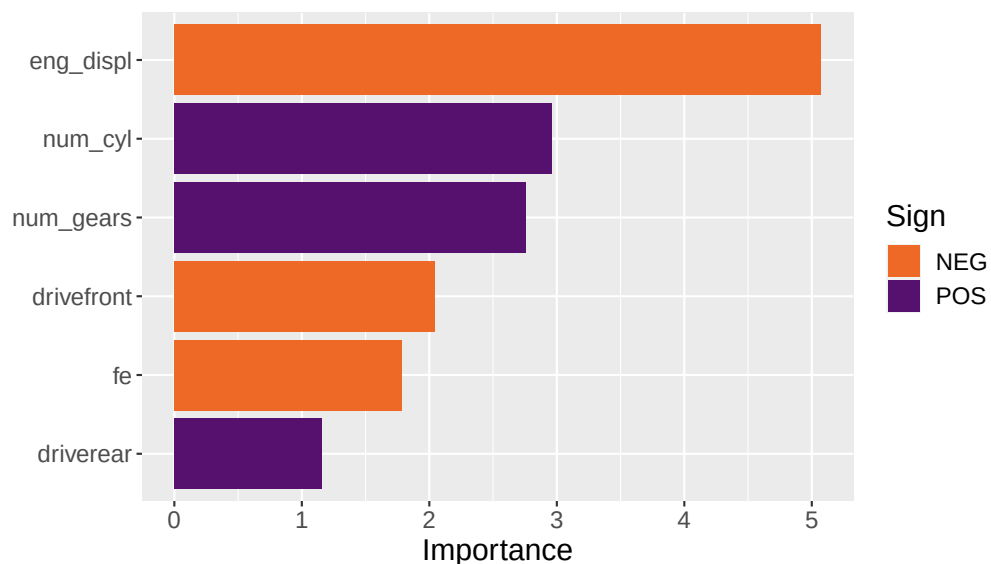
Please turn over for the questions...

20 marks

1. (a) A MARS model has been fitted to a random sample of the Hitters dataset. Using the following output, calculate the expected salary for a player with 63 runs batted in (`rbi`) and 9 home runs (`hm_run`). Please round your answer to the nearest 3 decimal places.

```
# Call: earth(formula=salary~rbi+hm_run, data=data,
#             keepxy=TRUE, nprune=~4)
#
# salary =
# 186.8413
# + 31.19324 * pmax(0, 22 - rbi)
# + 12.62587 * pmax(0, rbi - 22)
# - 18.8807 * pmax(0, hm_run - 13)
#
# Selected 4 of 13 terms, and 2 of 2 predictors (nprune=4)
# Termination condition: Reached nk 21
# Importance: rbi, hm_run
# Number of terms at each degree of interaction: 1 3
# (additive model)
# GCV 160972.1   RSS 29497933   GRSq 0.1987167
# RSq 0.2470239
```

- (b) A logistic regression model has been fitted to a random sample of the cars2011 dataset to predict whether a car's engine is "Turbocharged" (against the reference level of "Naturally Aspirated"). Using the variable importance plot below to answer the following questions:



- Which is the most important predictor in the model?
- Which is the most important positive predictor in the model?
- As `num_gears` increases, what do we expect will happen to the probability that our model predicts "Naturally Aspirated"?
- Is the importance value based on a z -statistic?

(c) (i) Consider the following equation:

$$E \left[\left(Y_0 - \hat{f}(x_0) \right)^2 \right] = \text{var} \left(\hat{f}(x_0) \right) + \left(E \left[\hat{f}(x_0) \right] - f(x_0) \right)^2 + \text{var}(\epsilon)$$

Which term in this equation is the bias?

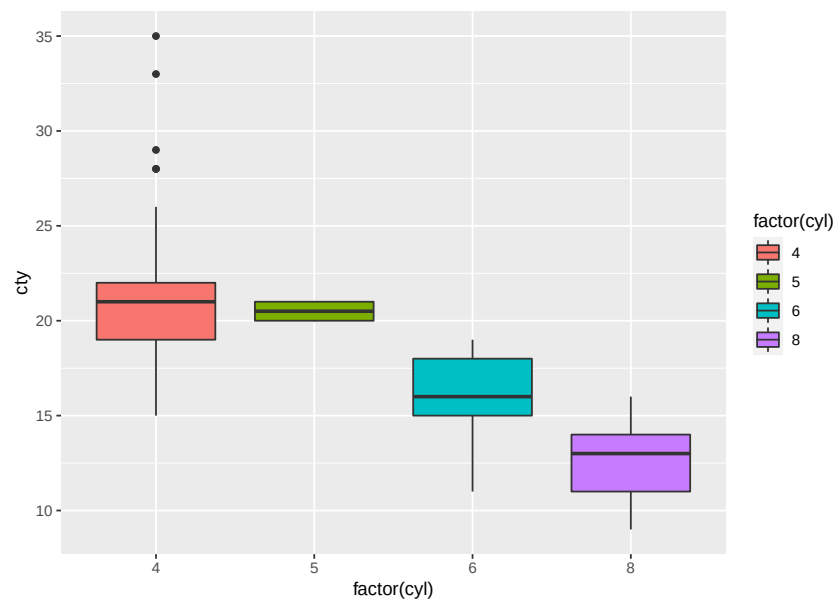
(ii) Consider the following `tidymodels` workflow in R:

```
# == Workflow =====  
# Preprocessor: Recipe  
# Model: logistic_reg()  
#  
# -- Preprocessor -----  
# 3 Recipe Steps  
#  
# * step_dummy()  
# * step_center()  
# * step_BoxCox()  
#  
# -- Model -----  
# Logistic Regression Model Specification (classification)  
#  
# Computational engine: glm
```

Which pre-processing steps does this workflow include?

- Centring
- Scaling
- Dealing with skewness
- Converting categorical variables to dummies
- Imputing categorical variables
- Imputing numeric variables

(iii) Consider the following plot:



Which of the following commands should be used to produce this plot in R?

- `ggplot(mpg, aes(displ, cty, col = factor(cyl))) + geom_point()`
- `ggplot(mpg, aes(cty, displ, col = factor(cyl))) + geom_point()`
- `ggplot(mpg, aes(factor(cyl), cty, fill = factor(cyl))) + geom_boxplot()`
- `ggplot(mpg, aes(factor(cyl), cty, col = factor(cyl))) + geom_boxplot()`

(iv) Consider following table in R:

```
# A tibble: 3 × 2
#   drv   'mean(hwy)'
#   <chr>      <dbl>
# 1 4         19.2
# 2 f         28.2
# 3 r         21
```

Which of the following commands could be used to produce this table?

- `mpg %>% group_by(cyl) %>% summarise(mean(cty))`
- `mpg %>% group_by(drv) %>% summarise(mean(cty))`
- `mpg %>% group_by(cyl) %>% summarise(mean(hwy))`
- `mpg %>% group_by(drv) %>% summarise(mean(hwy))`

20 marks

2. Consider the linear regression model where the intercept term $\beta_0 = 0$:

$$y_i = \beta x_i + \epsilon_i$$

with ϵ_i are i.i.d. $N(0, \sigma^2)$.

- (a) Derive $\hat{\beta}$, the least-squares estimate of β .
- (b) Show that the least-squares estimate is unbiased.
- (c) Derive the variance of $\hat{\beta}$, $\text{var}(\hat{\beta})$.

20 marks

3. There are four key skills in a data scientist's arsenal: analysis, mathematics, coding, and communication. In this question, we'll practice your communication skills. Whenever you are communicating, it's important to consider your audience. Different audiences will need to know different information, and be able to understand different levels of technical detail.

Write a brief report explaining data cleaning and how to apply it in `tidymodels`. Your report should explain:

- why data cleaning is important;
- what data cleaning entails; and
- how data cleaning is performed in `tidymodels`.

Marks will be allocated for the quality and correctness of your writing, as well as how suitable your writing is for the intended audience.

20 marks

4. An analysis of the Titanic dataset is given on the following pages. Use the analysis to answer the following questions. Justify your answers by referring to the relevant section(s) of the analysis.

(a) How many passengers are in the Titanic dataset?

(b) Consider the following R code:

```
titanic <-  
  titanic %>%  
  mutate(  
    across(where(is.character), factor)  
  )
```

What does this code do?

(c) What predictors are in the final model?

(d) What tuning parameters are tuned in the analysis?

(e) What are the best tuning parameter values?

(f) What is the AUC for the final model when fitted to the test dataset?

(g) What is the predicted probability of survival for a 3rd class 51 year old male travelling with his wife, but no parents or children?

Titanic analysis

Jono Tuke

Last compiled on 05 October, 2021

Load data

```
data("titanic_train", package = "titanic")
titanic <- as_tibble(titanic_train)
titanic
```

```
## # A tibble: 891 x 12
##   PassengerId Survived Pclass Name    Sex    Age SibSp Parch Ticket  Fare Cabin
##   <int>      <int>  <int> <chr>  <chr> <dbl> <int> <int> <chr>  <dbl> <chr>
## 1         1         0      3 Braun~ male   22     1     0 A/5 2~  7.25 ""
## 2         2         1      1 Cumin~ fema~  38     1     0 PC 17~ 71.3 "C85"
## 3         3         1      3 Heikk~ fema~  26     0     0 STON/~  7.92 ""
## 4         4         1      1 Futre~ fema~  35     1     0 113803 53.1 "C12~
## 5         5         0      3 Allen~ male   35     0     0 373450  8.05 ""
## 6         6         0      3 Moran~ male   NA     0     0 330877  8.46 ""
## 7         7         0      1 McCar~ male   54     0     0 17463  51.9 "E46"
## 8         8         0      3 Palss~ male    2     3     1 349909 21.1 ""
## 9         9         1      3 Johns~ fema~  27     0     2 347742 11.1 ""
## 10        10         1      2 Nasse~ fema~  14     1     0 237736 30.1 ""
## # ... with 881 more rows, and 1 more variable: Embarked <chr>
```

Clean data

```
titanic <- titanic %>% clean_names()
titanic
```

```
## # A tibble: 891 x 12
##   passenger_id survived pclass name      sex    age sib_sp parch ticket  fare
##   <int>      <int>  <int> <chr>    <chr> <dbl> <int> <int> <chr>  <dbl>
## 1         1         0      3 Braund, M~ male   22     1     0 A/5 2~  7.25
## 2         2         1      1 Cumings, ~ fema~  38     1     0 PC 17~ 71.3
## 3         3         1      3 Heikkinen~ fema~  26     0     0 STON/~  7.92
## 4         4         1      1 Futrelle,~ fema~  35     1     0 113803 53.1
## 5         5         0      3 Allen, Mr~ male   35     0     0 373450  8.05
## 6         6         0      3 Moran, Mr~ male   NA     0     0 330877  8.46
## 7         7         0      1 McCarthy,~ male   54     0     0 17463  51.9
## 8         8         0      3 Palsson, ~ male    2     3     1 349909 21.1
## 9         9         1      3 Johnson, ~ fema~  27     0     2 347742 11.1
## 10        10         1      2 Nasser, M~ fema~  14     1     0 237736 30.1
## # ... with 881 more rows, and 2 more variables: cabin <chr>, embarked <chr>
```



```
titanic <-
  titanic %>%
  mutate(
    across(where(is.character), factor)
  )
titanic
```

```
## # A tibble: 891 x 12
##   passenger_id survived pclass name      sex      age sib_sp parch ticket  fare
##   <int>      <int>   <int> <fct>    <fct> <dbl>   <int> <int> <fct>  <dbl>
## 1         1         0     3 Braund, M~ male    22      1     0 A/5 2~  7.25
## 2         2         1     1 Cumings, ~ fema~   38      1     0 PC 17~ 71.3
## 3         3         1     3 Heikkinen~ fema~   26      0     0 STON/~  7.92
## 4         4         1     1 Futrelle,~ fema~   35      1     0 113803 53.1
## 5         5         0     3 Allen, Mr~ male    35      0     0 373450  8.05
## 6         6         0     3 Moran, Mr~ male    NA      0     0 330877  8.46
## 7         7         0     1 McCarthy,~ male    54      0     0 17463  51.9
## 8         8         0     3 Palsson, ~ male     2      3     1 349909 21.1
## 9         9         1     3 Johnson, ~ fema~   27      0     2 347742 11.1
## 10        10         1     2 Nasser, M~ fema~   14      1     0 237736 30.1
## # ... with 881 more rows, and 2 more variables: cabin <fct>, embarked <fct>
```

```
titanic <-
  titanic %>%
  mutate(
    survived = factor(survived),
    pclass = factor(pclass)
  )
```

```
titanic <-
  titanic %>%
  select(survived, pclass, sex, age, sib_sp, parch)
titanic
```

```
## # A tibble: 891 x 6
##   survived pclass sex      age sib_sp parch
##   <fct>    <fct> <fct>  <dbl> <int> <int>
## 1 0        3     male    22      1     0
## 2 1        1     female  38      1     0
## 3 1        3     female  26      0     0
## 4 1        1     female  35      1     0
## 5 0        3     male    35      0     0
## 6 0        3     male    NA      0     0
## 7 0        1     male    54      0     0
## 8 0        3     male     2      3     1
## 9 1        3     female  27      0     2
## 10 1       2     female  14      1     0
## # ... with 881 more rows
```

Split data

```
titanic_split <- initial_split(titanic, strata = survived)
titanic_split
```

```
## <Analysis/Assess/Total>
```

```
## <667/224/891>
titanic_train <- training(titanic_split)
titanic_test <- testing(titanic_split)

titanic_cv <- vfold_cv(titanic_train, strata = survived)
titanic_cv

## # 10-fold cross-validation using stratification
## # A tibble: 10 x 2
##   splits          id
##   <list>         <chr>
## 1 <split [599/68]> Fold01
## 2 <split [600/67]> Fold02
## 3 <split [600/67]> Fold03
## 4 <split [600/67]> Fold04
## 5 <split [600/67]> Fold05
## 6 <split [600/67]> Fold06
## 7 <split [601/66]> Fold07
## 8 <split [601/66]> Fold08
## 9 <split [601/66]> Fold09
## 10 <split [601/66]> Fold10
```

Set up recipe

```
titanic_recipe <- recipe(survived ~ ., data = titanic)
```

Set up model

```
titanic_model <-
  decision_tree(
    mode = "classification",
    min_n = tune(),
    tree_depth = tune()
  ) %>%
  set_engine("rpart")
```

Set up workflow

```
titanic_wf <-
  workflow() %>%
  add_recipe(titanic_recipe) %>%
  add_model(titanic_model)
titanic_wf
```

```
## == Workflow =====
## Preprocessor: Recipe
## Model: decision_tree()
##
## -- Preprocessor -----
## 0 Recipe Steps
##
```

```
## -- Model -----
## Decision Tree Model Specification (classification)
##
## Main Arguments:
##   tree_depth = tune()
##   min_n = tune()
##
## Computational engine: rpart
```

Set up grid

```
titanic_grid <- grid_regular(min_n(), tree_depth())
titanic_grid
```

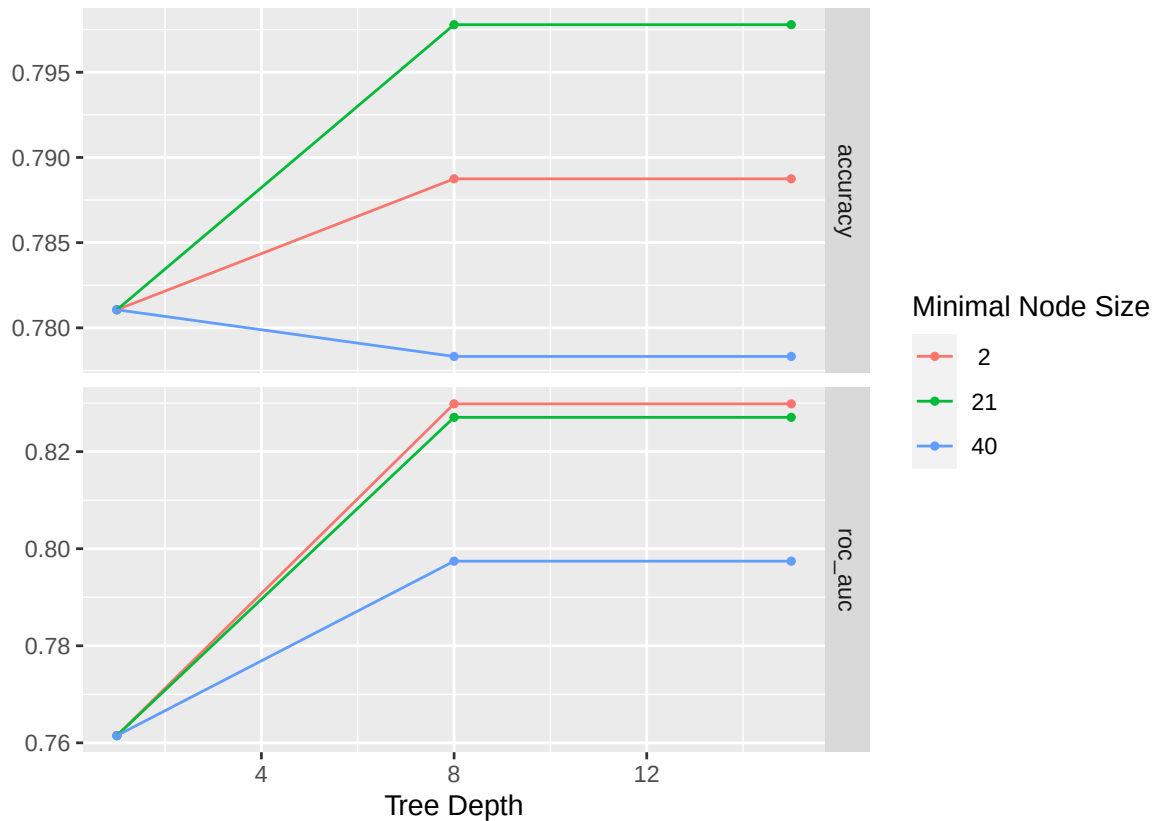
```
## # A tibble: 9 x 2
##   min_n tree_depth
##   <int>     <int>
## 1     2         1
## 2    21         1
## 3    40         1
## 4     2         8
## 5    21         8
## 6    40         8
## 7     2        15
## 8    21        15
## 9    40        15
```

Tune model

```
doParallel::registerDoParallel()
titanic_tune <- tune_grid(
  titanic_wf,
  resamples = titanic_cv,
  grid = titanic_grid,
  control = control_resamples(save_pred = TRUE)
)
```

Choose model

```
titanic_tune %>% autoplot()
```



```
show_best(titanic_tune, metric = "roc_auc")
```

```
## # A tibble: 5 x 8
##   tree_depth min_n .metric .estimator   mean     n std_err .config
##   <int> <int> <chr>   <chr>     <dbl> <int>   <dbl> <chr>
## 1      8      2 roc_auc binary    0.830    10  0.0204 Preprocessor1_Model14
## 2     15      2 roc_auc binary    0.830    10  0.0204 Preprocessor1_Model17
## 3      8     21 roc_auc binary    0.827    10  0.0214 Preprocessor1_Model15
## 4     15     21 roc_auc binary    0.827    10  0.0214 Preprocessor1_Model18
## 5      8     40 roc_auc binary    0.797    10  0.0205 Preprocessor1_Model16
```

```
titanic_wf <-
  titanic_wf %>%
  finalize_workflow(select_best(titanic_tune, metric = "roc_auc"))
```

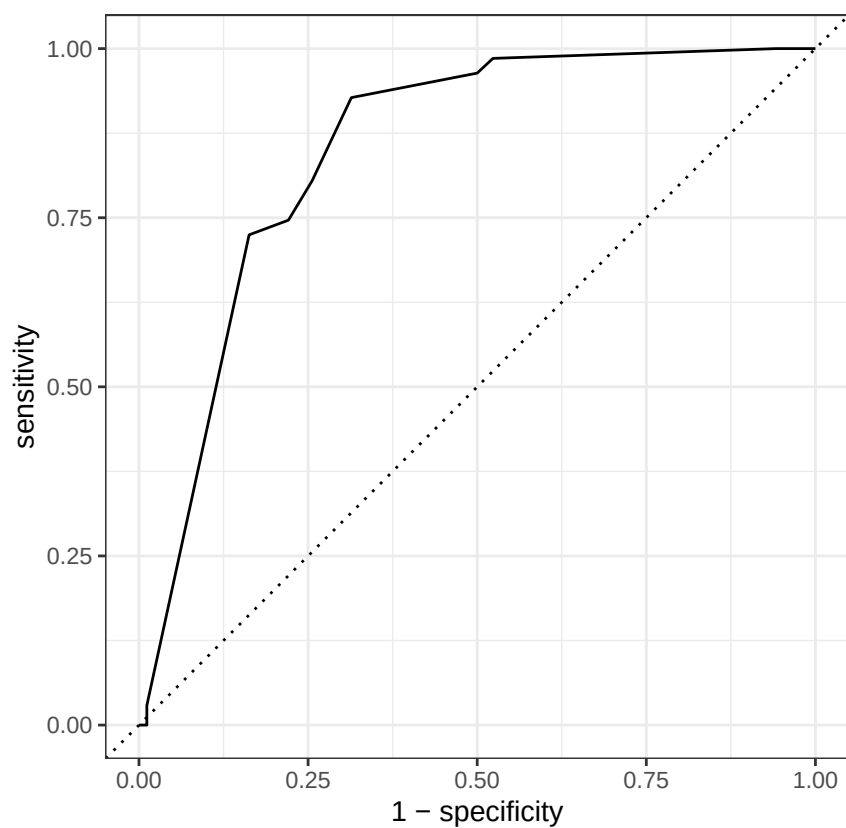
Quick last fit to see how works on test dataset.

```
titanic_LF <- titanic_wf %>% last_fit(titanic_split)
titanic_LF %>% collect_metrics()
```

```
## # A tibble: 2 x 4
##   .metric .estimator .estimate .config
##   <chr>   <chr>         <dbl> <chr>
## 1 accuracy binary         0.835 Preprocessor1_Model1
## 2 roc_auc  binary         0.849 Preprocessor1_Model1
```

```
titanic_LF %>%
  collect_predictions() %>%
  roc_curve(truth = survived, estimate = .pred_0) %>%
```

```
autoplot()
```



Prediction

```
newdata <- tibble(  
  pclass = "3",  
  sex = "male",  
  age = 51,  
  sib_sp = 1,  
  parch = 0  
)  
titanic_LF %>%  
  extract_fit_parsnip() %>%  
  predict(new_data = newdata)
```

```
## # A tibble: 1 x 1  
##   .pred_class  
##   <fct>  
## 1 0
```

```
titanic_LF %>%  
  extract_fit_parsnip() %>%  
  predict(new_data = newdata, type = "prob")
```

```
## # A tibble: 1 x 2  
##   .pred_0 .pred_1
```

##	<dbl>	<dbl>
## 1	0.891	0.109